III B.Tech – II Semester

Course Code	**COST EFFECTIVE HOUSING TECHNIQUES** **Professional Elective-III**	L	T	P	C
23A01605a		3	0	0	3

Course Objectives (COs):

The objectives of this course are to make the student to:

1. Analyze the housing scenarioin urban and rural areas, including challengesin housing finance and urban planning.
2. Explore and evaluate innovative low-cost housing technologies for sustainable construction.
3. Investigate alternative building materials and infrastructure services for cost-effective housing solutions.
4. Assess rural housing techniques, including traditional mud housing, soil stabilization, and fire treatment for roofing.
5. Develop strategies forhousingin disaster-prone areas, with a focus on earthquake, cyclone, and flood-resistant construction.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Examine the current status of urban and rural housing and analyze the role of finance and planningin housing development.
2. Evaluate and recommend cost-effective construction techniques, including prefabrication and innovative roofing/flooring systems.
3. Assess the feasibility of alternative building materials and infrastructure solutions for low-cost housing.
4. Analyze traditional rural housing methods and propose modern techniques for improving rural housing quality.
5. Design housing solutions for disaster-prone areas by incorporating earthquake, cyclone, and flood-resistant strategies.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	1	2	-	2	2	-	-	1	-	1	2	2
CO -2	3	3	2	3	2	-	3	-	-	-	-	2	3	2
CO -3	3	2	3	3	2	-	3	-	-	-	-	2	3	3
CO -4	2	3	3	3	2	2	2	-	-	-	-	2	3	3
CO -5	2	3	3	3	3	3	3	-	-	-	-	2	3	3

UNIT – I

a) **Housing Scenario :**Introducing - Status of Urban Housing - Status of Rural Housing

b) **Housing Finance**: Introducing - Existing Finance System in India - Government Role as Facilitator - Status At Rural Housing Finance – ImpedimentlyIn Housing Finance and Related Issues

c) **Land Use and Physical Planning for Housing :**Introduction - Planning of Urban Land - Urban Land Ceiling and Regulation Act - Efficiency of Building Bye Lass - Residential Densities

d) **Housing The Urban Poor :**Introduction - Living Conditions in Slums - Approaches and Strategies for Housing Urban Poor.

UNIT – II		
Development and adoption of low cost Resilient housing technology Introduction - Adoption of Innovative Cost Effective Construction Techniques - Adoption of Precast Elements in Partial Prefabrication- Adopting of total Prefabrication of Mass Housing in India- General Remarks On Pre Cast Rooting/Flooring Systems -Economical Wall System - Single Brick Thick Loading Bearing Wall - 19cm Thick Load Bearing Masonry Walls - Half Brick Thick Load Bearing Wall – Fly-Ash Gypsum Thick for Masonry - Stone Block Masonry - Adoption of Precast R.C. Plank and Join System for Roof/Floor in The Building		
UNIT – III		
Alternative Building Materials for Low Cost Housing Introduction - Substitute for Scarce Materials – Ferro-Cement - Gypsum Boards - Timber Substitutions - Industrial Wastes - Agricultural Wastes - Alternative Building Maintenance **Low Cost Infrastructure Services:** Introduction - Present Status - Technological Options - Low Cost Sanitation - Domestic Wall - Water Supply, Energy		
UNIT – IV		
Rural Housing: Introduction Traditional Practice of Rural Housing Continuous - Mud Housing Technology Mud Roofs - Characteristics of Mud - Fire Treatment for Thatch Roof - Soil Stabilization - Rural Housing Program		
UNIT – V		
Housing in Disaster prone areas: Introduction – Earthquake - Damages to Houses - Traditional Prone Areas - Type of Damages and Railways of Non-Engineered Buildings - Repair and Restore Action of Earthquake Damaged Non-Engineered Buildings Recommendations for Future Constructions. Requirement'sOf Structural Safety of Thin Precast Roofing Units Against Earthquake forces Status of R&D in Earthquake Strengthening Measures - Floods, Cyclone, Future Safety		
TEXT BOOKS:		
1. Building materials for low – income houses – International council for building research studies and documentation. 2. Hand book of low cost housing by A.K.Lal – Newage international publishers. 3. Low cost Housing – G.C. Mathur by South Asia Books		
REFRENCE BOOKS:		
1. Properties of concrete – Neville A.m. Pitman Publishing Limited, London. 2. Light weight concrete, Academic Kiado, Rudhai.G – Publishing home of Hungarian Academy of Sciences 1963. 3. Modern trendsin housing in developing countries – A.G. MadhavaRao, D.S. Rama chandra Murthy &G.Annamalai. E. & F. N. Spon Publishers		
Online Learning Resources:		
https://nptel.ac.in/courses/124107001		